INCH-POUND
A-A-59556
16 June 2000
SUPERSEDING
MIL-S-24035E
19 May 1988

COMMERCIAL ITEM DESCRIPTION

SAWS, BAND, CUTOFF, VERTICAL, TILT-FRAME, TRAVELING

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers metal-cutting manual and automatic tilt-frame vertical band saws used for mitered cutoff and notching operations on solid metal bars, structural shapes, extrusions, castings, pipes, and tubes.

2. CLASSIFICATION. The band saws shall be of the following types and sizes. The type and size shall be furnished as specified (see 7.2(b)). The size classification refers to the maximum bar stock capacity the machine can accommodate.

Type I - Manual stock indexing
Type II - Automatic stock indexing

Size 1 - 10 by 10 inches
Size 2 - 14 by 21 inches
Size 3 - 18 by 20 inches
Size 4 - 20 by 20 inches
Size 5 - 25 by 25 inches
Size 6 - 25 by 28 inches

3. SALIENT CHARACTERISTICS

3.1 General requirements. The band saw shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3405

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:18Z
Check the source to verify that this is the current version before use.

3.2 Components. The band saw shall, as a minimum, have the following components:

3.2.1 Frame. The frame shall be constructed with the strength and rigidity to meet the performance requirements of section 3.3. The frame shall consist of the saw blade drive wheel, saw blade idler wheel, drive motor, saw blade, and blade guides. The frame shall provide means for blade tensioning. The frame shall be capable of vertically tilting 45 degrees left and right for mitered cuts. A graduated degree scale shall be provided on both sides of the saw frame or a digital display showing the angle in 1 degree increments ranging from 0 to 45 degrees. A means shall be provided to securely lock the angle of cut. The frame base shall have a means to level and anchor the machine. Access doors or covers shall be provided for blade changes, adjustments, and servicing. When specified (see 7.2(c)), the tilting operation shall be hydraulically powered.

3.2.2 Traveling frame. A powered traveling frame mechanism shall be provided that drives the entire saw frame into the workpiece. Means shall be provided to limit the distance of both forward and return motions of the saw frame.

3.2.3 Blade drive motor. The blade drive motor shall be variable speed.

3.2.4 Automatic stock feed. Type II band saws shall have an automatic stock feeding device incorporated into the saw. The device shall allow the length and number of cuts to be preset. The machine cycle shall consist of the cut, saw frame retract, and stock index to the next position to begin the subsequent cut. The automatic stock feeding device shall consist of a powered feed and the manufacturer's standard number of powered material vises (see section 3.2.5) for the machine specified. When specified (see 7.2(d)), a discharge table shall be furnished with the automatic feeding device.

3.2.5 Material holding vise. The material holding vise(s) shall be constructed to securely hold workpieces while performing both vertical and mitered cuts. The material holding vise for the type I band saws shall be quick-acting and manually operated. Type II band saws shall have automatic hydraulic vises with adjustable clamping force.

3.2.6 Guidepost. An adjustable blade guidepost shall be provided and shall contain the upper blade guides. The guidepost shall be vertically adjustable to accommodate the maximum workpiece thickness required. The blade guides shall consist of carbide-faced roller bearings or zero clearance carbide faced guides with flow through coolant and shall be constructed with the strength and rigidity to meet the cutting requirements shown in table I. The blade guides shall provide the required 90 degrees twist in the saw blade.

3.2.7 Blade cleaner. A wiper, fixed brush, or powered wire brush for removing chips from the blade teeth shall be provided. When specified, a manual or automatic chip conveyor (see 7.2(e)) and chip tray (see 7.2(f)) shall be provided.

3.2.8 Saw blades. When specified (see 7.2 (g)), saw blades of the specified material, quantity, size, tooth pitch and type shall be furnished with the band saw.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:18Z
Check the source to verify that this is the current version before use.

3.2.9 Controls. All controls shall be convenient to the operator. The control system shall consist of push buttons and selector switches for start/stop, emergency stop, blade speed, cutting force, and blade tension. When specified (see 7.2(h)), the type II band saw shall be provided with a numerical control (NC) or programmable logic controller (PLC). The automatic control shall allow the length and number of cuts to be programmed from a control console. The capability to set cutting pressures and blade speeds at the control console shall also be provided. If the machine is provided with NC or PLC control, programming functions and operational data entry shall be provided through a manual data input (MDI) system with an alphanumeric feedback display. The control system shall have the capability to store multiple cutting programs that can be individually retrieved.

3.2.10 Coolant system. A coolant system consisting of a sump, power driven pump, and coolant nozzle(s) shall be provided. Means shall be provided to determine and maintain coolant level within the sump. A removable strainer shall be provided to remove metal chips from the coolant.

3.2.11 Hydraulic system. When furnished, the hydraulic system shall be provided with a fluid level indicator and overpressure protection.

3.2.12 Electrical system. The band saw electrical system shall conform to National Fire Protection Association (NFPA) 79, "Electrical Standard for Industrial Machinery" (DoD adopted). Unless otherwise specified (see 7.2(i)), the machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(j)). The control system power shall be provided by a separate 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

3.2.13 Electrical motors. When furnished, the band saw electric drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1, "Motors and Generators" (DoD adopted). Unless otherwise specified (see 7.2(k)), the motor(s) shall have dripproof enclosures.

3.2.14 Optional equipment. Optional equipment shall be furnished as specified (see 7.2(l)).

3.3 Performance. Unless otherwise specified (see 7.2(m)), the band saw shall be capable of producing cuts in Society of Automotive Engineering (SAE) 1010 steel that comply with the requirements shown in table I when tested in accordance with sections 5.3.1 through 5.3.3 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:18Z
Check the source to verify that this is the current version before use.

TABLE I. Performance requirements.

Size	Minimum cutting rate (square inches per minute)	Perpendicularity of cut (inch per inch of diameter)	Maximum angular error at 45 degrees (degree)
1	11	±0.002	±0.5
2	11	±0.002	±0.5
3	11	±0.002	±0.5
4	16	±0.002	±0.5
5	16	±0.003	±0.5
6	16	±0.003	±0.5

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(n)).

3.5 Safety and health requirements. The band saw shall be designed and manufactured in accordance with ANSI B11.10, "Metal Sawing Machines - Safety Requirements for Construction, Care and Use" (DoD adopted). The manufacturer shall ensure the machine and all equipment and accessories used on the machine shall comply with Occupational Safety and Health Administration's (OSHA) Title 29 of the Code of Federal Regulations (CFR), Part 1910, "Occupational Safety and Health Standards". If a conflict arises between the OSHA and ANSI standards, the OSHA standards shall apply.

3.6 System of units. The US Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(o)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(p)), measuring and indicating devices shall be graduated in both the US and SI systems of measurement.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(q)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer's name
c. Machine model designation
d. Power input (volts, total amperes, phase, frequency)
e. Short-circuit/over-current rating
f. Contract number or order number
g. National stock number
h. Date of manufacture

3.8 Lubrication plate or chart. When specified (see 7.2(r)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application
b. Servicing interval
c. Type of lubricant(s) with SAE number or lubricant identifier

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:18Z
Check the source to verify that this is the current version before use.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The band saw shall meet all applicable Environmental Protection Agency restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Inspection. The band saw shall be inspected for compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(s)), the band saw shall be tested in accordance with sections 5.3.1 through 5.3.3.

5.3.1 Test conditions. The band saw shall be tested under the normal environmental conditions of the manufacturer's facility.

5.3.2 Operational test. The band saw shall be operated in accordance with the manufacturer's standard operating procedures. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls and features shall be verified during the trial period. For the tests described herein, the choice of saw blade, coolant, blade speed and force shall be the option of the supplier.

5.3.3 Performance tests. The band saw shall undergo a straight cut test. The test workpiece for machine sizes 1 through 3 shall have a diameter of 6 inches. The test workpiece for machine sizes 4 through 6 shall have a diameter of 10 inches. Cuts shall be made in both the vertical and 45 degrees tilted positions. The test results shall be compared to the performance requirements shown in table I.

5.4 Acceptance. Unless otherwise specified (see 7.2(t)), preliminary and final acceptance testing shall be conducted at the manufacturer's site. The results of the performance tests will be compared to the performance requirements shown in table I. Failure of the machine to meet the requirements shown in table I for the performance category selected shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:18Z
Check the source to verify that this is the current version before use.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the band saw shall be preserved, packed, and marked as specified in the acquisition order (see 7.2(u)).

7. NOTES

7.1 Sources of documents.

7.1.1 Government documents. Copies of CFR and FAR may be obtained from the U.S. Government Printing Office, Superintendent of Documents, Mail Stop: SSOP, Washington, DC 20402-9328. Electronic copies of CFR documents may be obtained from http://www.access.gpo.gov.

7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.1.3 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

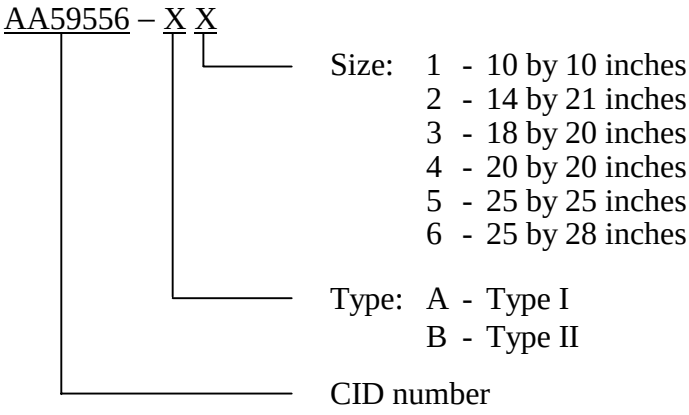
7.1.4 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.2 Ordering data. Acquisition documents must specify the following information:

a. CID document number, revision, and CID PIN
b. Type and size of vertical band saw required (see 2)
c. Hydraulically powered frame tilt, if required (see 3.2.1)
d. Discharge table, if required (see 3.2.4)
e. Chip conveyor, specify manual or powered, if required (see 3.2.7)
f. Chip tray, if required (see 3.2.7)
g. Saw blade material, quantity, size, tooth pitch and type, if required (see 3.2.8)
h. Numerical/programmable control system, if required (see 3.2.9)
i. Electrical requirements, if different (see 3.2.12)
j. Voltage, as specified (see 3.2.12)
k. Motor enclosure, if different (see 3.2.13)
l. Optional equipment, if required (see 3.2.14)
m. Performance requirements, if different (see 3.3)
n. Machine dimensions (length, width, and height) and weight, if different (see 3.4)
o. Specific measurement system, if required (see 3.6)
p. Dual graduation of indicating devices, if required (see 3.6)
q. Nameplate, if different (see 3.7)
r. Lubrication plate, if required (see 3.8)
s. Test requirements, if different (see 5.3)
t. Acceptance criteria, if different (see 5.4)
u. Packaging requirements (see 6.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:18Z
Check the source to verify that this is the current version before use.

7.3 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



MILITARY INTERESTS:

Custodians:
Army - AL
Navy - SH
Air Force - 99

Reviewer:
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

(Project 3405-0105)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:18Z
Check the source to verify that this is the current version before use.